

# Compact spatial continuation through the third transition

This component belongs to the complete local coupled-viscous-control proof. It uses the actual coefficient trajectory constructed in the third-growth entry and third-transition components, whose quantitative conclusions are restated explicitly below. Every additional spatial map, force, estimate and extension asserted here is proved in this component.

## 1 Full compact spatial realization through the third transition

We continue the actual third-growth fields through the entire third transition just proved. The second vorticity now revives with negative sign. Both older velocities and temperatures enter the spatial equations. The construction below uses the actual coefficient solution and proves its complete spatial realization, including a smooth finite extension of its forces. Its endpoint is the third transition endpoint; no third return is asserted here.

### 1.1 The original objects and a uniform material bound

The physical equations, coordinates, and orientation remain

$$\begin{aligned} \theta_t + u \cdot \nabla \theta &= d\Delta\theta + f_\theta, & x &\in \mathbb{R}^2, \\ u_t + (u \cdot \nabla)u &= d\Delta u + \theta e_2 + f_u, & \operatorname{div} u &= 0, \quad p = 0, \\ J &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, & e(s) &= (\sin s, \cos s). \end{aligned} \quad (1)$$

Here  $d > 0$  is the common physical viscosity and thermal diffusivity;  $A_0 \geq 1$ ,  $\nu = \sqrt{A_0}$ , and  $\mu = \lambda_0 = \lceil \nu/2 \rceil$  retain their original values. In particular  $\nu$  is a time scale. Retain  $K_j = \mu \hat{\lambda}_j$ , the exact source integer recursion, the source step  $\eta$ , and the actual scales  $\sigma_1, \sigma_2$ . Put

$$t_b = t_2^b, \quad t_a = t_3^a, \quad s = s_3 = L_3\sigma_1/\sigma_2, \quad \Gamma = \Gamma_3 = \sigma_2 \sin s, \quad t_1 = t_a + \Gamma^{-1}, \quad I = [t_b, t_1]. \quad (2)$$

These are the actual second return, third threshold, and third transition endpoint already constructed. Their proved bounds include  $0 < t_a - t_b < 14/\sigma_1$ ,  $L_3 \geq 16384$ ,  $0 < s \leq a := \sin s_2 \leq 1/512$ , and  $\Gamma \geq (15/16)L_3\sigma_1$ . Thus

$$|I| = (t_a - t_b) + \Gamma^{-1} < 15/\sigma_1. \quad (3)$$

All earlier activation weights are one on  $I$ ; the third is  $w_3(t) = \eta(\Gamma(t - t_b))$  and is one before  $t_a$ . The exact source control on  $I$  is  $\dot{\alpha} = (1 - \lambda)\mathcal{F}_2 + \lambda\mathcal{F}_3$ , where  $\lambda(t) = \eta(\Gamma(t - t_a))$  is zero during growth. The two switches  $w_3, \lambda$  have different original start times.

Use the determinant coordinates of the third-transition theorem:

$$\begin{aligned} h_0 &= \cos s_2, \quad D = h^2 + a^2, \quad r_b = |\zeta_2(t_b)|, \quad b = r_b \sin s, \quad R = |\zeta_3|^2 = (g^2 + b^2)/D, \\ \zeta_1 &= \frac{(h \sin \vartheta - a \cos \vartheta, h \cos \vartheta + a \sin \vartheta)}{\sqrt{D}}, \\ \zeta_2 &= \sqrt{D}(\sin \vartheta, \cos \vartheta), \\ \zeta_3 &= \frac{(g \sin \vartheta + b \cos \vartheta, g \cos \vartheta - b \sin \vartheta)}{\sqrt{D}}, \quad \vartheta = su. \end{aligned} \quad (4)$$

The symbol  $D$  in this display is a scalar, not a matrix. The actual growth and transition estimates give, on all of  $I$ ,

$$\begin{aligned}
h_0 \leq h < 13/4, \quad 1 \leq D < 11, \quad 3/4 < g < 5, \quad e^{-1/64} \leq R < 3, \\
0 \leq u \leq 640a/L_3 < 1/1024, \quad \cos \vartheta > 15/16, \quad g \cos \vartheta - b \sin \vartheta > 2/3, \\
0 < \Omega_1 < 10\sigma_1, \quad -7040a\sigma_2/L_3 \leq \Omega_2 \leq 0, \quad 0 < -\Theta_1 \leq 2\sigma_1^2/K_1, \\
0 < -\Theta_2 \leq 3\sigma_2^2/K_2, \quad 0 < -\Theta_3 \leq e^4 P_3^*, \quad |\Omega_3| \leq 3e^4 K_3 P_3^*/\sigma_2, \\
P_3^* = (A_0/\mu) \hat{\lambda}_3^{-7/8}, \quad 1 \leq r_b \leq \sqrt{10}.
\end{aligned} \tag{5}$$

For the bound on  $R$ , growth has  $1 \leq R \leq 2$ , and the transition changes  $\log |\zeta_3|$  by less than  $1/128$ . The amplitude bounds follow from the actual growth target, the transition gain at most four, and the proved quotient at most three. They are consequences of the constructed trajectory, not additional conditions imposed on an unknown flow.

For completeness the exact material equations for all three layers are

$$\begin{aligned}
G_{<j} &= -A_0 R_\alpha e_2 + \sum_{i < j} w_i K_i \Theta_i \zeta_i, & D_{<j} &= \dot{\alpha} J + \sum_{i < j} \frac{w_i \Omega_i}{r_i^2} J \zeta_i \otimes \zeta_i, \\
\dot{M}_j &= D_{<j} M_j, & M_j(t_{0j}) &= I, \quad \zeta_j = M_j^{-T} p_j, \quad r_j = |\zeta_j|, \\
\dot{\Theta}_j &= -\frac{J \zeta_j \cdot G_{<j}}{K_j r_j^2} \Omega_j - d K_j^2 r_j^2 \Theta_j, & \dot{\Omega}_j &= K_j \zeta_{j,1} \Theta_j - d K_j^2 r_j^2 \Omega_j.
\end{aligned} \tag{6}$$

Here  $p_1 = e(s_*)$ ,  $p_2 = e(s_2)$ ,  $p_3 = e(s)$ , and  $t_{03} = t_b$ ; the first two original insertion times are unchanged. The scalar  $r_j^2$  is strictly positive. Each matrix  $D_{<j}$  has trace zero, since  $\zeta_i \cdot J \zeta_i = 0$ . The exact matrices are

$$\begin{aligned}
M_1 &= R_\alpha, \quad M_2 = R_{\alpha-s_*} \begin{pmatrix} 1 & -(h-h_0)/a \\ 0 & 1 \end{pmatrix}, \\
\mathcal{L}(t) &= (\zeta_2(t) \ \zeta_3(t)), \quad M_3 = \mathcal{L}(t)^{-T} \mathcal{L}(t_b)^T, \quad M_3^{-1} = \mathcal{L}(t_b)^{-T} \mathcal{L}(t)^T.
\end{aligned} \tag{7}$$

Indeed  $\zeta_2$  is annihilated by the transpose of its own added shear. Thus both columns of  $\mathcal{L}$  solve  $\dot{z} = -D_{<3}^T z$ . Their determinant is  $\det \mathcal{L} = -b$ , including at birth. Differentiating the last two expressions proves  $\dot{M}_3 = D_{<3} M_3$ ,  $M_3(t_b) = I$ , and  $M_3^{-T} p_3 = \zeta_3$ . The same calculation, or the displayed triangular form, proves the first two material equations and their original data. All matrices have determinant one, and hence define global linear diffeomorphisms of the original  $\mathbb{R}^2$ .

Let  $\|\cdot\|$  be Euclidean operator norm and  $\|\cdot\|_F$  Frobenius norm. For a two by two matrix its adjugate has the same Frobenius norm: it permutes its four entries and changes two signs. Consequently

$$\|\mathcal{L}^{-1}\| \leq \|\mathcal{L}\|_F/b, \quad \|\mathcal{L}\|_F^2 = D + R < 14, \quad \|\mathcal{L}(t_b)\|_F^2 = r_b^2 + 1 \leq 11.$$

Applying these inequalities to both products in (7) gives the uniform bounds

$$\begin{aligned}
N_1 &= 1, \quad N = 1 + (3 - h_0)/a, \quad N_2 = \tilde{N} = 1 + (13/4 - h_0)/a \leq 2N, \\
N_3 &= \mathcal{M} = 16/\sin s, \quad \|M_j\|, \|M_j^{-1}\| \leq N_j \quad (1 \leq j \leq 3).
\end{aligned} \tag{8}$$

For  $j = 3$  the product bound is  $\sqrt{154}/b < 16/\sin s$ . For  $j = 2$ , write its triangular factor as  $I - q e_1 \otimes e_2$  and its inverse as  $I + q e_1 \otimes e_2$ , so both norms are at most  $1 + |q|$ . Finally  $2N - \tilde{N} = 1 + (11/4 - h_0)/a > 0$ . The third estimate is valid when  $\Omega_2$  is nonzero; it does not use the shorter growth-only shear formula.

## 1.2 The exact source supports remain nested

The birth bound  $P_{2,b} \leq e^{18}(A_0/\mu)\hat{\lambda}_2^{-7/8}$  gives  $A_{2,b} = K_2 r_b P_{2,b} \leq \sqrt{10}e^{18}\nu^2\hat{\lambda}_2^{1/8}$ . The already proved scale relation  $\sigma_2^2 \leq 5A_{2,b}/4$  therefore gives

$$\begin{aligned} C_2 &= e^9 \sqrt{5\sqrt{10}/4}, & \sigma_2 &\leq C_2 \nu \hat{\lambda}_2^{1/16}, \\ \sigma_1 &\geq \sqrt{e} \nu \hat{\lambda}_1^{1/16}, & C_M &= \frac{256C_2}{15\sqrt{e}}, & \mathcal{M} &\leq C_M \hat{\lambda}_1^{(Q_2-1)/16}. \end{aligned} \quad (9)$$

For the last estimate use  $\sin s \geq 15L_3\sigma_1/(16\sigma_2)$ ,  $L_3 \geq 1$ , and the unchanged recursion  $\hat{\lambda}_2 = \hat{\lambda}_1^{Q_2}$ . With  $N \leq C_N \hat{\lambda}_1^{1/16}$ , the strengthened actual source selection includes the strict inequalities

$$\hat{\lambda}_1^{47Q_2/16-3} > 4C_N C_M, \quad \hat{\lambda}_1^{(31Q_2+1)/16} > \sqrt{11}C_M/s_0. \quad (10)$$

Their exponents are positive because  $Q_2 \geq 202$ . Multiplication by the corresponding negative frequency powers gives

$$\begin{aligned} 2N_2 N_3 \hat{\lambda}_2^{-3} &< \hat{\lambda}_1^{-3}, & \sqrt{11}N_3 \hat{\lambda}_2^{-2} &< s_0, \\ N_2 \hat{\lambda}_1^{-3} &< s_0/(2C_\ell), & N_2 \hat{\lambda}_1^{-2} &< s_0. \end{aligned} \quad (11)$$

For the first, the exponent of  $\hat{\lambda}_1$  in  $4C_N C_M \hat{\lambda}_1^{Q_2/16-3Q_2+3}$  is  $-(47Q_2/16-3)$ . For the second its exponent is  $(Q_2-1)/16-2Q_2 = -(31Q_2+1)/16$ . The last two follow from the earlier actual first-source margins  $N\hat{\lambda}_1^{-3} < s_0/(4C_\ell)$  and  $N\hat{\lambda}_1^{-2} < s_0/2$ , together with  $N_2 \leq 2N$ . Thus the doubling needed in the third transition is absorbed by proved strict margins.

Retain the original profile, primitive, and spatial cutoffs. Namely  $F(s) = \sin s + \chi_F(s)(s - \sin s)$  on  $[-\pi, \pi]$  is extended periodically;  $\chi_F$  is even, smooth, supported away from the endpoints, and equals one on  $[-s_0, s_0]$ ,  $0 < s_0 < 1$ . The even periodic primitive is uniquely specified by  $P' = F$  and  $\int_{-\pi}^{\pi} P = 0$ . On the affine cell it is  $P(s) = P(0) + s^2/2$ , with its original constant  $P(0)$  retained. The radial  $f \in C_c^\infty(B_1)$  equals one on  $B_{1/2}$ . Keep the exact source radii and material envelopes

$$\begin{aligned} \ell_1 &= s_0/(C_\ell \mu), & \ell_2 &= \hat{\lambda}_1^{-3}/\mu, & \ell_3 &= \hat{\lambda}_2^{-3}/\mu, & C_\ell &\geq 4, \\ g_j(x, t) &= f(M_j^{-1}x/\ell_j), & k_j &= K_j \zeta_j, & \xi_j &= k_j \cdot x, & \rho_j &= |k_j|^2. \end{aligned} \quad (12)$$

The vectors  $k_j$  here are physical phases, not the source integer seed exponents. Every third field and derivative is supported in  $|x| \leq N_3 \ell_3$ . At such points

$$|M_2^{-1}x| \leq N_2 N_3 \ell_3 < \ell_2/2, \quad |K_2 \zeta_2 \cdot x| \leq K_2 \sqrt{11} N_3 \ell_3 < s_0.$$

Thus the full second field is affine on an open neighborhood of the third support. Every point of the second support satisfies  $|x| \leq N_2 \ell_2 < \ell_1/2$  and  $|K_1 \zeta_1 \cdot x| < s_0$  by the last two inequalities in (11);  $M_1$  is a rotation. The first support lies in  $B_{\ell_1}$ , where the original radial base cutoff is one and  $|\mu R_\alpha e_2 \cdot x| < s_0$ . This proves every inclusion in the original laboratory coordinates. Strictness ensures an open affine neighborhood for every support and every derivative support, not only an identity at its center.

## 1.3 All fields, original initial data, and full forces

Retain the original radial  $\chi_0 \in C_c^\infty(B_{R_0})$ , equal to one on  $B_2$ , and the radial  $H_0 \in C_c^\infty(B_{R_H})$ ,  $R_H > R_0 + 1$ , with  $H_0 = |x|^2/2$  on  $B_{R_0+1}$ . Define

$$\begin{aligned} \theta_{\text{in}} &= -\frac{A_0}{\mu} \sin(\mu x_2) \chi_0, & \theta_{\text{ref}} &= -\frac{A_0}{\mu} F(\mu x_2) \chi_0, \\ \theta_b &= -\frac{A_0}{\mu} F(\mu R_\alpha e_2 \cdot x) \chi_0 + (1 - \eta(\nu t))(\theta_{\text{in}} - \theta_{\text{ref}}), & \psi_b &= \dot{\alpha} H_0, & u_b &= J \nabla \psi_b. \end{aligned} \quad (13)$$

The angle is zero before  $1/\nu$ . Thus this is exactly the original sine initial datum and zero initial velocity. The interpolation has finished before the first insertion and hence before  $I$ . For  $j = 1, 2, 3$  define the active increments

$$\begin{aligned} T_j &= w_j \Theta_j, \quad Q_j = w_j \Omega_j / \rho_j, \quad V_j = T_j F(\xi_j) g_j, \quad \Psi_j = Q_j P(\xi_j) g_j, \quad v_j = J \nabla \Psi_j, \\ \theta &= \theta_b + \sum_{j=1}^3 V_j, \quad \psi = \psi_b + \sum_{j=1}^3 \Psi_j, \quad u = J \nabla \psi. \end{aligned} \quad (14)$$

For  $t \leq t_{0j}$  the  $j$ th increment is zero. Flatness of the original  $\eta$  makes this zero extension smooth. Before  $t_b$  these formulas give the earlier constructed spatial solution exactly. On  $I$  they use the actual revived  $\Omega_2$  and its corresponding streamfunction; none of the older fields is frozen.

For each layer suppress its index temporarily and set  $k = k_j$ ,  $\rho = \rho_j$ ,  $\xi = k \cdot x$ ,  $g = g_j$ ,  $D_* = D_{<j}$ ,  $G_* = G_{<j}$ . All profiles in the next formulas are evaluated at  $\xi$ . Set

$$Q_r = \frac{\dot{w}\Omega + w k_1 \Theta}{\rho} - Q \frac{\dot{\rho}}{\rho}, \quad \dot{\rho} = -2k \cdot D_*^T k, \quad \dot{Q} = Q_r - d\rho Q. \quad (15)$$

The exact force increments, with no pressure reconstruction, are

$$\begin{aligned} E_{\theta,j} &= \dot{w} \Theta F g + Q P(J \nabla g \cdot G_*) + Q T(F^2 - P F') g(J k \cdot \nabla g) \\ &\quad - dT\{\rho(F + F'')g + 2F'k \cdot \nabla g + F \Delta g\}, \end{aligned} \quad (16)$$

$$E_{u,j} = 2D_* v + Q_r J(k F g + P \nabla g) + (\nabla v)v - T F g e_2 - d(\rho v + \Delta v). \quad (17)$$

The complete spatial differentiations in these expressions are

$$\begin{aligned} v &= Q J(k F g + P \nabla g), \\ \nabla v &= Q J\{k \otimes k F' g + F(k \otimes \nabla g + \nabla g \otimes k) + P \nabla^2 g\}, \\ \Delta V &= T\{\rho F'' g + 2F'k \cdot \nabla g + F \Delta g\}, \\ \Delta v &= Q J\{\rho k F'' g + \rho F' \nabla g + 2F'k(k \cdot \nabla g) + 2F(\nabla^2 g)k + F k \Delta g + P \nabla \Delta g\}. \end{aligned} \quad (18)$$

Keep the full base residuals

$$\begin{aligned} f_{\theta,b} &= \nu \eta'(\nu t)(\theta_{\text{ref}} - \theta_{\text{in}}) - d \Delta \theta_b, \\ f_{u,b} &= \ddot{\alpha} J \nabla H_0 + \dot{\alpha}^2 (\nabla(J \nabla H_0)) J \nabla H_0 - \theta_b e_2 - d \dot{\alpha} J \nabla \Delta H_0, \\ f_{\theta} &= f_{\theta,b} + \sum_{j=1}^3 E_{\theta,j}, \quad f_u = f_{u,b} + \sum_{j=1}^3 E_{u,j}. \end{aligned} \quad (19)$$

*Verification of every equation and interaction.* On a  $j$ th support the proved nesting makes the full older fields exactly  $G_* \cdot x$  and  $D_* x$ . For  $\mathcal{D}_j = \partial_t + D_* x \cdot \nabla$ , differentiating  $M_j^{-1}$  gives  $\mathcal{D}_j(M_j^{-1}x) = 0$ . Likewise  $\dot{k} = -D_*^T k$  implies  $\mathcal{D}_j \xi = 0$ . Therefore  $\mathcal{D}_j g = 0$ ,  $\mathcal{D}_j V = \dot{T} F g$ , and  $\mathcal{D}_j \Psi = \dot{Q} P g$ . Direct contractions give

$$\begin{aligned} v \cdot G_* &= Q F g(J k \cdot G_*) + Q P(J \nabla g \cdot G_*), \\ v \cdot \nabla V &= Q T(F^2 - P F') g(J k \cdot \nabla g). \end{aligned}$$

The second identity uses  $J k \cdot k = 0$ ,  $J \nabla g \cdot \nabla g = 0$ , and  $J \nabla g \cdot k = -J k \cdot \nabla g$ . The coupling in  $\dot{\Theta}$  in (6) cancels exactly the first term of the first identity. Adding  $\dot{w} \Theta F g$  and subtracting the complete  $d \Delta V$  proves (16).

For an arbitrary trace-free two by two  $D_*$ , direct entrywise multiplication gives  $D_* J + J D_*^T = 0$ . Hence

$$\mathcal{D}_j(J \nabla \Psi) = D_* v + J \nabla(\mathcal{D}_j \Psi).$$

The reverse velocity interaction is  $(v \cdot \nabla)(D_*x) = D_*v$ . Adding it, the exact self-advection, and the negative buoyancy and diffusion terms gives (17), including its factor two. The product rule applied to  $\Psi = QPg$  proves all entries of (18); differentiating  $\Delta\Psi = Q(\rho F'g + 2Fk \cdot \nabla g + P\Delta g)$  in particular produces all six vector Laplacian groups.

These are full ordered interactions. In the third increment,  $\mathcal{D}_3$  contains both  $v_1 \cdot \nabla V_3$  and  $v_2 \cdot \nabla V_3$ , and both corresponding advections of  $v_3$ . The term  $v_3 \cdot G_{<3}$  contains both  $v_3 \cdot \nabla V_1$  and  $v_3 \cdot \nabla V_2$ ; the term  $D_{<3}v_3$  contains both reverse velocity interactions. The same statements for  $j = 2$  include all pairs of the first two layers, and  $j = 1$  includes the base. Thus adding in order 1, 2, 3 counts every ordered pair exactly once. Outside a layer's support its derivatives vanish; the identities consequently telescope on all of  $\mathbb{R}^2$ , not merely on an affine neighborhood.

The base radial cutoff is transported by  $\dot{\alpha}Jx$ , and so is the base phase  $\mu R_\alpha e_2 \cdot x$ . On its scalar support  $u_b = \dot{\alpha}Jx$ ; before  $1/\nu$  the angle and velocity are zero. This proves the scalar base formula, including the initial interpolation. Differentiating  $u_b = \dot{\alpha}J\nabla H_0$  proves its vector formula globally. The sum therefore solves (1) on  $[0, t_1]$  with the original data. Since  $u = J\nabla\psi$ , equality of mixed spatial partial derivatives gives divergence zero.  $\square$

The exact spatial coordinate correspondence is also retained. For each  $j$  put  $C_j = M_j^{-1}M_j^{-T}$  and  $\tilde{g}_j(y) = f(y/\ell_j)$ . On a function of independent variables  $(\xi, y)$ , pulled back by  $(\xi, y) = (K_j\zeta_j \cdot x, M_j^{-1}x)$ , the chain rule gives

$$\begin{aligned}\nabla_x &= K_j\zeta_j\partial_\xi + M_j^{-T}\nabla_y, \\ \Delta_x &= K_j^2r_j^2\partial_\xi^2 + 2K_j(C_jp_j \cdot \nabla_y)\partial_\xi + \sum_{l,m}(C_j)_{lm}\partial_{y_l}\partial_{y_m}.\end{aligned}\tag{20}$$

Multiplying the commuting constant spatial derivatives proves the second equality from the first; in particular it retains all mixed terms and the actual time-dependent metric.

On an affine cell  $g_j = 1$ ,  $|\xi_j| < s_0$ , the exact fields are

$$V_j = T_jK_j\zeta_j \cdot x, \quad v_j = \frac{w_j\Omega_j}{r_j^2}J\zeta_j \otimes \zeta_j x, \quad \Delta V_j = \Delta v_j = 0.\tag{21}$$

This follows by differentiating the original  $Q_j(P(0) + \xi_j^2/2)$ . The constant remains in the full cutoff field; it was not discarded from its definition. The damping forces in the core remain exactly  $-d\rho_jV_j$  and  $-d\rho_jv_j$ . On an uncut sine-profile region instead,  $F'' = -F$  gives  $\rho_jV_j + \Delta V_j = 0$  and  $\rho_jv_j + \Delta v_j = 0$ . All other vector residual terms in (17) remain. Thus the core and sine calculations give the exact map from the damped amplitude trajectories to the released profile, including the forces that the profile requires.

#### 1.4 Quantitative spatial costs for every layer and both forces

For any scalar, vector or tensor field  $U$ , write  $|U|_m = \sup_x \|D_x^m U\|$  in the Euclidean multilinear norm. Let  $[W]_q = \sup_\xi |W^{(q)}(\xi)|$  for a periodic profile and  $f_m = \sup_y \|D^m f(y)\|$ . Define fixed bounds on  $I$  by

$$\begin{aligned}r_{-,j}^2 &= (1, 1, 63/64)_j, & r_{+,j}^2 &= (1, 11, 3)_j, & \kappa_j &= K_j r_{+,j}, \\ U_{\Theta,j} &= (2\sigma_1^2/K_1, 3\sigma_2^2/K_2, e^4 P_3^*)_j, \\ U_{\Omega,j} &= (10\sigma_1, 7040a\sigma_2/L_3, 3e^4 K_3 P_3^*/\sigma_2)_j, \\ W_{j,1} &= (0, 0, \Gamma[\eta']_0)_j, & \vartheta_* &= 640as/L_3, & A_* &= 270a^2\sigma_1 + 6aU_{\Omega,2}\vartheta_*.\end{aligned}\tag{22}$$

The lower third length follows from  $e^{-1/64} \geq 1 - 1/64 = 63/64$ . To verify the control bound  $|\dot{\alpha}| \leq A_*$ , write

$$y = \frac{ag + hb}{D}, \quad \mathcal{F}_2 = \frac{a\Omega_1(h \tan \vartheta - a)}{D}, \quad \mathcal{F}_3 = \frac{\Omega_1 y (h \sin \vartheta - a \cos \vartheta) + b\Omega_2 \sin \vartheta}{g \cos \vartheta - b \sin \vartheta}.$$

On the proved box  $0 \leq h \tan \vartheta < a$ ,  $y < 18a$ ,  $b < 4a$ , and the denominator of  $\mathcal{F}_3$  exceeds  $2/3$ . Thus  $|\mathcal{F}_2| \leq 10a^2\sigma_1$  and  $|\mathcal{F}_3| \leq 270a^2\sigma_1 + 6aU_{\Omega,2}\vartheta_*$ . Their convex combination satisfies the asserted bound, including the preceding growth interval.

For each layer set the explicit constants

$$\begin{aligned} D_j^* &= A_* + \sum_{i < j} U_{\Omega,i}, & G_j^* &= A_0 + \sum_{i < j} K_i U_{\Theta,i} r_{+,i}, \\ Q_j^* &= \frac{U_{\Omega,j}}{K_j^2 r_{-,j}^2}, & Q_{r,j}^* &= \frac{W_{j,1} U_{\Omega,j} + \kappa_j U_{\Theta,j} + 2D_j^* U_{\Omega,j}}{K_j^2 r_{-,j}^2}. \end{aligned} \quad (23)$$

The rank-one operator  $J\zeta_i \otimes \zeta_i/r_i^2$  has norm one, proving  $\|D_{<j}\| \leq D_j^*$ . The gradient sum in (6) proves  $|G_{<j}| \leq G_j^*$ . The identity  $|\dot{\rho}_j/\rho_j| \leq 2D_j^*$  proves  $|Q_j| \leq Q_j^*$  and  $|Q_{r,j}| \leq Q_{r,j}^*$ . In particular the second-parent velocity contributes the complete  $U_{\Omega,2}$  to  $D_3^*$ , with no sign-based removal.

Define finite sums

$$\begin{aligned} e_{j,m} &= (N_j/\ell_j)^m f_m, \\ B_{j,m}(W) &= \sum_{q=0}^m \binom{m}{q} \kappa_j^q [W]_q e_{j,m-q}, \\ C_{j,m}(P) &= \sum_{q=0}^m \binom{m}{q} \kappa_j^q [P]_q e_{j,m-q+1}. \end{aligned} \quad (24)$$

The  $q$  derivatives allocated to the linear phase have size at most  $\kappa_j^q$ ; the other  $m - q$  derivatives of the linear material coordinate have size at most  $(N_j/\ell_j)^{m-q}$ . The product rule with all  $\binom{m}{q}$  allocations therefore proves  $|g_j|_m \leq e_{j,m}$ ,  $|W(\xi_j)g_j|_m \leq B_{j,m}(W)$ , and  $|P(\xi_j)\nabla g_j|_m \leq C_{j,m}(P)$ . These bounds hold for every choice of unit derivative directions.

The full damping-plus-diffusion costs are

$$\begin{aligned} \mathfrak{D}_{\theta,j,m} &= dU_{\Theta,j} \{ \kappa_j^2 B_{j,m}(F) + 2B_{j,m+2}(F) \}, \\ \mathfrak{D}_{u,j,m} &= dQ_j^* \{ \kappa_j^2 B_{j,m+1}(P) + 2B_{j,m+3}(P) \}. \end{aligned} \quad (25)$$

They bound, respectively,  $-d(\rho_j V_j + \Delta V_j)$  and  $-d(\rho_j v_j + \Delta v_j)$ . The explicit factor two bounds the sum of the two coordinate traces defining the Laplacian; it is not an identification with a single second derivative. The complete force estimates are

$$\begin{aligned} |E_{\theta,j}|_m &\leq W_{j,1} U_{\Theta,j} B_{j,m}(F) + Q_j^* G_j^* C_{j,m}(P) + \mathfrak{D}_{\theta,j,m} \\ &\quad + Q_j^* U_{\Theta,j} \sum_{q=0}^m \binom{m}{q} B_{j,q+1}(P) B_{j,m-q+1}(F), \end{aligned} \quad (26)$$

$$\begin{aligned} |E_{u,j}|_m &\leq (2D_j^* Q_j^* + Q_{r,j}^*) B_{j,m+1}(P) + U_{\Theta,j} B_{j,m}(F) + \mathfrak{D}_{u,j,m} \\ &\quad + (Q_j^*)^2 \sum_{q=0}^m \binom{m}{q} B_{j,q+1}(P) B_{j,m-q+2}(P). \end{aligned} \quad (27)$$

For the scalar nonlinear term apply the product rule to  $v_j \cdot \nabla V_j$ ; for the vector term apply it to  $(\nabla v_j)v_j$ . The displayed derivative allocations give exactly the two sums. All other terms follow individually from (16)–(18). This proves the estimates with activation, all parent interactions, self-advection, buoyancy, moving phase length, and diffusion included.

For the base on  $I$ , let  $c_m = \sup_x \|D^m \chi_0(x)\|$ ,  $H_m = |J\nabla H_0|_m$ , and

$$B_{0,m}(F) = \sum_{q=0}^m \binom{m}{q} \mu^q [F]_q c_{m-q}.$$

Write  $A_{**}$  for the explicit bound on  $|\tilde{\alpha}|$  constructed in (34) below. Then

$$\begin{aligned} |f_{\theta,b}|_m &\leq 2d(A_0/\mu)B_{0,m+2}(F), \\ |f_{u,b}|_m &\leq A_{**}H_m + A_*^2 \sum_{q=0}^m \binom{m}{q} H_{q+1}H_{m-q} + (A_0/\mu)B_{0,m}(F) + 2dA_*H_{m+2}. \end{aligned} \quad (28)$$

These follow directly from (19), since the interpolation is complete on  $I$ . Adding these bounds to (26) and (27) gives explicit bounds for both total forces at every spatial order. The previously proved formulas cover  $[0, t_b]$  with the same initial interpolation; the maximum of its bounds and these new ones covers  $[0, t_1]$ . Every bound on  $I$  integrates with the explicit length  $(t_a - t_b) + \Gamma^{-1} < 15/\sigma_1$  from (3).

In particular all localization and frequency factors remain visible:

$$\begin{aligned} N_1/\ell_1 &= C_\ell \mu/s_0, \quad N_2/\ell_2 = \tilde{N} \mu \hat{\lambda}_1^3, \quad N_3/\ell_3 = \frac{16\mu \hat{\lambda}_2^3}{\sin s}, \\ dU_{\Theta,3}K_3^2 &= de^4 A_0 \mu \hat{\lambda}_3^{9/8}, \quad dQ_3^*K_3^3 = \frac{64e^4 dA_0 \mu \hat{\lambda}_3^{9/8}}{21\sigma_2}. \end{aligned} \quad (29)$$

These are exact factors in upper costs, not lower bounds for the total force. The exact modal diffusion exponents on  $I$  are

$$dK_1^2|I|, \quad dK_2^2 \int_I D dt, \quad dK_3^2 \int_I R dt. \quad (30)$$

The second lies between  $dK_2^2|I|$  and  $11dK_2^2|I|$ . The third splits exactly into the proved growth cost and  $\delta_3 \int_0^1 R d\tau$ , whose transition part lies between  $(63/64)\delta_3$  and  $3\delta_3$ , with  $\delta_3 = dK_3^2/\Gamma$ . Older coupling terms remain in (6); they have not been replaced by damping.

### 1.5 An explicit recursion for every temporal and mixed cost

We give the coefficient recursion as well as the spatial chain rule, so no stationary-envelope convention enters a time derivative. Use  $\tau = \Gamma(t - t_a)$ ,  $L = L_3$ ,  $q = \sigma_1/\sigma_2 = s/L$ ,  $\epsilon = \sigma_1/\Gamma$ ,  $\delta_j = dK_j^2/\Gamma$ , and the dimensionless state

$$\begin{aligned} z &= (h, g, u, E, F, \omega, V, X, Y), \\ E &= -K_1\Theta_1/\sigma_1^2, \quad F = -K_2\Theta_2/\sigma_2^2, \quad \omega = \Omega_1/\sigma_1, \quad V = \Omega_2/\sigma_2, \\ X &= -\Theta_3/P_3^*, \quad Y = -\sigma_2\Omega_3/(K_3P_3^*). \end{aligned} \quad (31)$$

The two last coordinates are signed linear amplitudes; forming them does not divide by a temperature. Put  $\bar{c} = A_0 \cos s_*/\sigma_1^2$  and  $\bar{C} = A_0 \sin s_*/\sigma_1^2$ , and define

$$\begin{aligned} x_c &= (hg - ab)/D, \quad y_c = (ag + hb)/D, \\ N_3 &= \sigma_1^2(E + \bar{c})y_c + \sigma_1^2\bar{C}x_c + b\sigma_2^2F, \\ \mathcal{K} &= \frac{N_3}{\sigma_2^2 \sin s R}, \quad \mathcal{B} = \frac{g \sin(su) + b \cos(su)}{\sqrt{D} \sin s}. \end{aligned}$$

Let  $\Phi(z, \lambda) = (1 - \lambda)\mathcal{F}_2 + \lambda\mathcal{F}_3$ , with the exact feedbacks displayed above and substitutions  $\Omega_1 = \sigma_1\omega$ ,  $\Omega_2 = \sigma_2V$ ,  $\vartheta = su$ . The complete explicit dimensionless vector field  $\mathcal{V}(z, \lambda)$  is

$$\begin{aligned}
h' &= a\epsilon\omega, & g' &= \epsilon\omega \frac{2ahg + (h^2 - a^2)b}{D} + \frac{b}{\sin s}V, \\
u' &= \frac{-\Phi - a^2\sigma_1\omega/D}{\Gamma_s}, & E' &= -\epsilon\bar{C}\omega - \delta_1E, \\
F' &= -q\epsilon \frac{a(E + \bar{c}) + \bar{C}h}{D}V - \delta_2DF, & \omega' &= -\epsilon E \frac{h \sin(su) - a \cos(su)}{\sqrt{D}} - \delta_1\omega, \\
V' &= -F\sqrt{D} \frac{\sin(su)}{\sin s} - \delta_2DV, & X' &= KY - \delta_3RX, \\
Y' &= BX - \delta_3RY.
\end{aligned} \tag{32}$$

Every prime in this display is  $d/d\tau$ . Substitution of (31) into the physical equations proves each row, including both factors  $q\epsilon$  in the second-temperature row. Conversely those substitutions reverse, giving the physical pair, phases, and feedback. This explicitly proves the coordinate map and its inverse wherever the displayed denominators are positive.

For bounds let  $\mathcal{Z}$  be the closed rectangular box given by

$$\begin{aligned}
h &\in [h_0, 13/4], \quad g \in [3/4, 5], \quad u \in [0, 1/1024], \\
E &\in [1/8, 2], \quad F \in [1/16, 3], \quad \omega \in [0, 10], \\
V &\in [-1/32, 0], \quad X \in [0, e^4], \quad Y \in [0, 3e^4].
\end{aligned}$$

The proved actual solution on  $I$  belongs to this box. All feedback denominators are bounded away from zero on the whole box by the same elementary inequalities used above. In particular  $D \geq 1$  and  $g \cos(su) - b \sin(su) > 2/3$  there. Introduce independent symbols  $\lambda^{[0]}, \lambda^{[1]}, \dots$  for the jets of the control switch and define on each finite jet expression

$$\mathcal{C} = \sum_{l=1}^9 \mathcal{V}_l(z, \lambda^{[0]}) \partial_{z_l} + \sum_{r \geq 0} \lambda^{[r+1]} \partial_{\lambda^{[r]}}. \tag{33}$$

The second sum is finite on every expression to which it is applied. The chain rule proves that the physical  $r$ th time derivative of any state function  $H(z, \lambda)$  is  $\Gamma^r \mathcal{C}^r H$  evaluated at  $\lambda^{[j]} = \eta^{(j)}(\Gamma(t - t_a))$ . Before the transition these are zero for every  $j \geq 0$ ; the same identity therefore holds on all of  $I$ . Consequently the explicit control constants are

$$A_{r+1} = \Gamma^r \max \left\{ |\mathcal{C}^r \Phi| : z \in \mathcal{Z}, \quad 0 \leq \lambda^{[0]} \leq 1, \quad |\lambda^{[j]}| \leq [\eta^{(j)}]_0 \quad (1 \leq j \leq r) \right\}, \quad A_{**} = A_2. \tag{34}$$

This is a maximum of a fully specified continuous elementary function over an explicit compact rectangle; it is finite and depends only on the retained parameters and profile derivative bounds. It is not a supremum over an unspecified later solution. The equality of derivatives just proved yields  $|\alpha^{(r+1)}| \leq A_{r+1}$  at every order. The sharper  $A_*$  in (22) bounds its first derivative.

For a complete algebraic mixed-jet evaluation, set  $A_j = M_j^{-1}/\ell_j$  and introduce independent  $(t, x, \xi, y)$ . Define

$$\mathfrak{T}_j = \partial_t + (\dot{k}_j \cdot x) \partial_\xi + (\dot{A}_j x) \cdot \nabla_y, \quad \mathfrak{X}_{j,i} = \partial_{x_i} + k_{j,i} \partial_\xi + \sum_l (A_j)_{li} \partial_{y_l}. \tag{35}$$

For  $\iota_j^* H = H(t, x, k_j(t) \cdot x, A_j(t)x)$ , the ordinary chain rule, iterated one derivative at a time, gives

$$\partial_t^n \partial_x^\gamma \iota_j^* H = \iota_j^* (\mathfrak{T}_j^n \mathfrak{X}_{j,1}^{\gamma_1} \mathfrak{X}_{j,2}^{\gamma_2} H). \tag{36}$$

The commutators vanish: the time derivative of  $k_{j,i}$  cancels the  $x_i$  derivative of  $\dot{k}_j \cdot x$ , and the two corresponding terms for  $A_j$  cancel as well. Thus the identity also verifies the consistency of all mixed derivative orders. Every coefficient jet in this recursion is evaluated by (32)–(34) and

$$\dot{M}_j = D_{<j} M_j, \quad \dot{A}_j = -A_j D_{<j}, \quad \dot{k}_j = -D_{<j}^T k_j,$$

together with product differentiation of (6) and the exact activation jets  $\Gamma^r \eta^{(r)}$  for  $w_3$ . For example  $\partial_t^{n+1} A_j = -\sum_{q=0}^n \binom{n}{q} (\partial_t^q A_j) (\partial_t^{n-q} D_{<j})$ ; the other two linear equations have the identical product recursion with their displayed order and signs. Their zeroth bounds are (8), so induction gives explicit finite matrix and phase jet bounds at every prescribed order.

To obtain an actual numerical expression for a mixed norm, expand (36) on every displayed field or residual, keeping its finitely many summands. Replace each coefficient jet by the preceding recursive bound, each  $x^\beta$  by  $(N_j \ell_j)^{|\beta|}$  on its layer support, each periodic derivative by  $[F]_r$  or  $[P]_r$ , and each envelope derivative by  $f_m$ . Sum the absolute products with their explicit binomial coefficients. The triangle and product inequalities prove the resulting bound. For the base use its support radius  $R_H$  and its displayed functions instead. This procedure computes all mixed costs, including every time derivative of  $d\rho_j$  and the changing Laplacian metric; its recursion has finite depth  $n + |\gamma|$ .

## 1.6 The transition endpoint and a proved smooth force collar

At  $t_1$  all weights are one,  $\Theta_3(t_1) < 0$  and  $\Omega_3(t_1) < 0$ , with the exact inverse amplitudes of the transition theorem. In the common affine cell the complete state is

$$\theta = \left( -A_0 R_\alpha e_2 + \sum_{j=1}^3 K_j \Theta_j \zeta_j \right) \cdot x, \quad u = \left( \dot{\alpha} J + \sum_{j=1}^3 \frac{\Omega_j}{r_j^2} J \zeta_j \otimes \zeta_j \right) x. \quad (37)$$

The second vorticity has the revived negative sign proved above; the first remains positive. Formula (37) and the full cutoff formulas give actual states of (1), with their exact material matrices and laboratory gravity.

To continue smoothly to the right, use the explicit vector field (32) with the constant control value  $\lambda = 1$ . Let  $z_1 = z(t_1)$  and take the closed cube  $\mathcal{C}_1 = \{z : \|z - z_1\|_\infty \leq 1/100\}$ . Its interior is in the smooth domain of this vector field. Indeed  $h > h_0 - 1/100 > 3/4$ ,  $g > 3/4 - 1/100 > 7/10$ , and  $|u| < 1/1024 + 1/100 < 1/50$  there. Since  $s \leq a \leq 1/512$  and  $b < 4s$ , these imply  $\cos(su) > 15/16$  and

$$g \cos(su) - b \sin(su) > (7/10)(15/16) - 4s^2/50 > 3/5.$$

Thus  $D$ ,  $R$ ,  $\sin s$  and every feedback denominator remain strictly positive on a neighborhood of the cube. No amplitude quotient occurs in its vector field. Define the concrete finite maxima

$$B = \max_{\mathcal{C}_1} \|\mathcal{V}(z, 1)\|_\infty, \quad L_{\text{lip}} = \max_{\mathcal{C}_1} \|D_z \mathcal{V}(z, 1)\|_{\infty \rightarrow \infty}, \quad (38)$$

$$\Delta\tau = \min \left\{ \frac{1}{400(1+B)}, \frac{1}{2(1+L_{\text{lip}})} \right\} > 0.$$

On continuous paths in this cube beginning at  $z_1$ , the Picard map  $z(\tau) \mapsto z_1 + \int_1^\tau \mathcal{V}(z(r), 1) dr$  moves points by at most  $B\Delta\tau < 1/400 < 1/100$ . The mean-value formula on the convex cube makes its Lipschitz constant at most  $L_{\text{lip}}\Delta\tau < 1/2$  in the uniform norm. Successive iterates therefore converge uniformly: their successive differences are bounded by a geometric series with ratio below  $1/2$ . The limit satisfies the integral equation, and the same contraction inequality proves uniqueness. Differentiation of this equation first gives  $C^1$ , and repeated differentiation of the smooth vector field gives  $C^\infty$ . This constructs an actual right collar of physical length

$$\epsilon_R = \Delta\tau/\Gamma > 0. \quad (39)$$

At  $t_1$  the original switching profile is flat and equal to one, so the control and every derivative match this continuation. The matrices and phase fields continue by their exact linear ODEs. Their determinant stays one by the trace identity, or by the explicit formulas with  $\det \mathcal{L} = -b$ . The angle continues by  $\dot{\alpha} = \mathcal{F}_3$ . Hence the fields have a smooth right extension with the actual endpoint jets, not a frozen state.

On this collar define the forces by the raw full residuals

$$f_\theta^{\text{raw}} = \theta_t + u \cdot \nabla \theta - d\Delta \theta, \quad f_u^{\text{raw}} = u_t + (u \cdot \nabla)u - d\Delta u - \theta e_2. \quad (40)$$

They agree with (19) on the entire required slab by the proof above. Thus no additional nesting assertion outside that slab is needed for the collar. They are smooth and compact in space, since the fields are smooth compact fields and their matrices vary continuously on a compact time interval. All collar mixed costs are explicitly computed by the same derivation and pullback recursions, replacing  $\mathcal{Z}$  by  $\mathcal{C}_1$  and  $\lambda$  by one in their maxima, and using the raw expressions (40). The compact matrix bounds follow directly from (7) on that cube; for example  $h < 13/4 + 1/100$ ,  $g < 5 + 1/100$ , and its positive denominators bound  $\mathcal{L}$  and  $\mathcal{L}^{-1}$  there. These explicit finite maxima provide every derivative order without assuming the original pre-endpoint amplitude bounds on the collar.

Extend the original sine state constantly to negative times, with zero velocity and inactive increments, and put  $\epsilon_L = 1/\nu$ . Use the raw residuals of these extensions and set

$$\beta(t) = \eta(1 + t/\epsilon_L)\eta(1 + (t_1 - t)/\epsilon_R), \quad \tilde{f}_\bullet = \beta f_\bullet^{\text{raw}}. \quad (41)$$

They equal the proved forces on  $[0, t_1]$ . Flatness of  $\eta$  makes their zero extension outside  $[-\epsilon_L, t_1 + \epsilon_R]$  smooth. Their support is compact in space and time. More explicitly, with

$$B_{\beta,r} = \sum_{q=0}^r \binom{r}{q} \epsilon_L^{-q} \epsilon_R^{-(r-q)} [\eta^{(q)}]_0 [\eta^{(r-q)}]_0,$$

the product rule gives for every multi-index  $\gamma$

$$\|\partial_t^n \partial_x^\gamma \tilde{f}_\bullet\|_\infty \leq \sum_{q=0}^n \binom{n}{q} B_{\beta,q} \|\partial_t^{n-q} \partial_x^\gamma f_\bullet^{\text{raw}}\|_\infty. \quad (42)$$

The right-hand mixed norms have the explicit finite recursion just proved. The PDE is asserted on  $[0, t_1]$ , where the cutoff is one.

Finally, the streamfunction is compact and smooth; hence the velocity has finite energy. If  $N_0(x) = (2\pi)^{-1} \log|x|$ , then  $N_0 * \Delta\psi = (\Delta N_0) * \psi = \psi$  by distributional integration by parts, and therefore  $u = J\nabla N_0 * \omega$ ,  $\omega = \Delta\psi$ , exactly. Odd  $F$ , even  $P$ , and even material envelopes make  $\theta, u, f_\theta, f_u$  odd and  $\omega$  even. Every field and force on the required slab has the original spatial support inside  $B_{R_H}$  by the proved nesting. This completes the full compact spatial extension through the actual third transition with every retained physical diffusion cost and both older interaction directions.